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The 26 August 2026 Chhochen Khola ice-rock avalanche–debris flow cascade on the Nepal–China border: dynamics, numerical computation, and causes

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Abstract: On 26 August 2026, a large ice–rock slides initiated in the Chhochen Khola catchment of the Nepal Himalaya and evolved into a high-speed debris flow that travelled down the transboundary Lende Khola river, crossing the Nepal–China border. This cascading geohazard devastated China's Gyirong Port, 22.8 km downstream, as well as villages and infrastructure farther downstream in Nepal. By integrating remote sensing, seismological, hydrological, and field evidence, we reconstruct a three-stage cascade. (1) A 0.76 km² ice–rock mass failed on the north flank of the Langtang Lirung massif at ~5,190 m a.s.l.. The initial slide had a total volume of ~103 Mm³, of which ~12.6 Mm³ (≈12%) was ice. (2) After ~30 s of the descent, the slide struck and entrained a detached glacier tongue of 0.92 km² at ~4,010 m a.s.l., generating a seismic signal equivalent to an Mw 5.1 event, and rapidly developed into a large ice–rock avalanche. Within the following ~100 s, ~67

Mm^3 of debris was deposited in the Chhochen Khola, whereas the remaining ~36 Mm^3 was remobilized as a highly mobile debris flow and entered the Lende Khola. (3) The debris flow travelled 14.2 km along the Lende Khola in ~350 s. At a bedrock-gorge cross-section near Gyirong Port, it attained a peak discharge of 2.46×10^5 m^3 s^{-1} with a cross-sectional mean velocity of 36.2 m s^{-1}. We simulated the ice–rockslides and the debris flows using a one-dimensional depth-averaged Voellmy model and a Saint-Venant shallow-water model, respectively. The simulations yield a maximum slide velocity of 129.2 m s^{-1} in the first stage and show that the peak discharge at the Gyirong section had attenuated to ~11% of that at the confluence of the Chhochen Khola and the Lende Khola. Meteorological and seismic records preceding the event reveal no single extreme rainfall or temperature anomaly, nor any recent tectonic trigger; rather, the disaster was a compound event arising from the combined effects of long-term climate warming, glacier retreat, adverse geological preconditioning, and freeze–thaw cycling.

Keywords: Rockslide; Ice avalanche; Debris flow; Hazard cascade; Transboundary disaster; Nepal Himalaya

1 Introduction

High-mountain catchments connect steep rock walls, glaciers, permafrost-affected terrain, and deeply incised rivers within short horizontal distances. A rock or ice failure high on a slope can therefore interact with snow, glacier ice, water, and sediment before reaching inhabited valleys far below its source (*Hu et al., 2025*; *Mani et al., 2023*; *Shugar et al., 2021*; *Zhang et al., 2022*). The Himalayan setting intensifies this connection: large relief provides substantial potential energy, while transport routes, hydropower works, ports, and settlements occupy narrow river corridors. Glacier thinning and retreat, rising snowline altitudes, and changes in frozen-ground conditions can alter the stability of steep source areas, although the timing and mechanism of any individual failure require event-specific evidence (Haeberli et al., 2017; Huggel et al., 2012).

Past disasters show how a moving mass can change character and consequences along its path. The 2015 Langtang avalanche combined snow, ice, and rock released during the Gorkha earthquake and buried a settlement in a confined Himalayan valley (*Fujita et al., 2017*; *Gnyawali et al., 2020*). In 2021, the Chamoli rock–ice collapse evolved into a highly mobile downstream debris flow, damaging infrastructure more than 20 km from its source(*Shugar et al., 2021*). Large glacier detachments on the Tibetan Plateau likewise demonstrate the mobility of ice-bearing failures in high-elevation terrain (*Kääb et al., 2018*). Together, these cases establish that source size and total runout alone do not describe the downstream hazard. The fraction of material deposited in intermediate valleys, the location of ice and water inputs, and the shape of the channel determine how much moving material reaches a particular settlement.

The physical evolution of a cascade is closely tied to the landscape it traverses. Impact and fragmentation can change grain-size distribution; entrainment can add ice, water, and channel sediment; and a decrease in gradient can promote deposition while a confined gorge can sustain rapid transport (*Hungr and Evans, 2004*; *Mani et al., 2023*; *Wang et al., 2024*). These processes need not occur at one sharp rheological threshold. They can overlap over a reach, with locally deposited material and a mobile fraction present at the same time. For that reason, stage boundaries inferred from imagery, deposits, and abrupt slope changes should be checked against independent timing evidence. A source-to-downstream reconstruction must also distinguish the speed of a flow front, a reach-averaged travel speed, and the propagation speed of a modeled discharge peak.

Numerical studies have moved from depth-averaged landslide runout models with prescribed resistance laws toward multiphase descriptions that represent interacting solid and fluid material, entrainment, and deposition (*Hungr and McDougall, 2009*; *Mergili et al., 2017*). Two-phase calculations reproduced aspects of the multistage 2015 Langtang avalanche (*Gnyawali et al., 2020*), while simulations of Chamoli combined remotely sensed geometry and event timing to explore the passage of rock–ice material into a destructive downstream flow (*Shugar et al., 2021*). *Zhang et al. (2023)* used RAMMS and

seismic/video constraints to analyze Chamoli's long-runout chain and test the influence of entrainment and model parameters. For channelized downstream transport, conservative one-dimensional Saint-Venant formulations provide a separate basis for estimating discharge, peak arrival, and hydrograph attenuation from cross-sectional geometry and an inlet hydrograph (*Hodges, 2019*). These studies show both the value of process-specific computation and the need to anchor each model in observations appropriate to the reach being represented.

On 26 August 2026, an ice–rock slide began on the north flank of the Langtang Lirung massif at approximately 5,190 m a.s.l. It struck a detached glacier tongue below the source and developed into an ice–rock avalanche through Chhochen Khola. Part of the material accumulated in the valley; the remaining mass entered the transboundary Lende Khola as a debris flow. The cascade reached Gyirong Port, approximately 22.8 km from the source, in about 8 min and continued downstream into Nepal. The separation between material stored in Chhochen Khola and material delivered to the main river makes this case especially useful for examining how valley geometry and material redistribution shape downstream hazard. The Chhochen Khola event offers a rare combination of satellite imagery, field evidence, seismic peaks, mapped deposits, and downstream hydraulic constraints, but the observations are uneven across its three process stages. The steep initial descent has a mapped source and a timed glacier impact, which support a longitudinal slide calculation. The intermediate ice–rock avalanche has clear impact and depositional evidence, yet its evolving ice, water, and sediment fractions are not measured along the valley. At the Lende Khola confluence, an estimated transferred volume, surveyed cross sections, and a downstream gorge estimate support a channel-routing calculation. We therefore use a Voellmy representation for the initial ice–rock slide and a Saint-Venant representation for the downstream debris flow, while reconstructing the intervening ice–rock avalanche from observed timing and deposits. This combination connects source failure to downstream discharge without assigning a single flow law to the entire cascade.

Here we integrate pre- and post-event remote sensing, field observations, topography, seismic timing, meteorological and cryospheric records, and stage-specific numerical calculations. We map source and deposit geometry, reconstruct travel distances and timing for the ice–rock slide, ice–rock avalanche, and debris flow, and estimate the material volume supplied to the Lende Khola. We then quantify the evolution of the debris-flow hydrograph toward Gyirong Port and examine the immediate and longer-term conditions associated with source-slope failure. The resulting reconstruction links the observed changes in process and mass distribution to the intensity and timing of the downstream flow.

2 Study Area

The Lende Khola is a transboundary river in the central Himalaya along the Nepal–China border and joins the Gyirong River near Gyirong Port, an important cross-border transport corridor (Fig. 1a-b). Its drainage area is approximately 897.96 km². The upper basin contains glacially dominated U-shaped valleys, whereas the lower basin is characterized by deeply incised fluvial gorges. Glaciers occupy approximately 166.87 km² of the upper basin (Zhang et al., 2026) (Fig. 1b). The Chhochen Khola catchment lies on the southern Himalayan slope and drains into the Guoba Xiaqu before entering the lower Lende Khola. Its headwaters include Tsangbu Ri and Langtang Lirung, with elevations exceeding 7,000 m, while the catchment outlet is near 2,910 m a.s.l. Glaciers cover approximately 10.75 km² of the catchment according to RGI 7.0. Exposed bedrock ridges, talus slopes, glacier ice, and permafrost-affected terrain occur in the source region. Downstream, the Lende Khola cuts metamorphic and granitic bedrock and alternates between confined V-shaped gorges and locally wider valley sections (Fig. 1c) (Zhang et al., 2026). The regional climate is controlled by the Indian summer monsoon, with most precipitation occurring from June to September; precipitation varies substantially with elevation and local topography (Immerzeel et al., 2014; Shu et al., 2025).

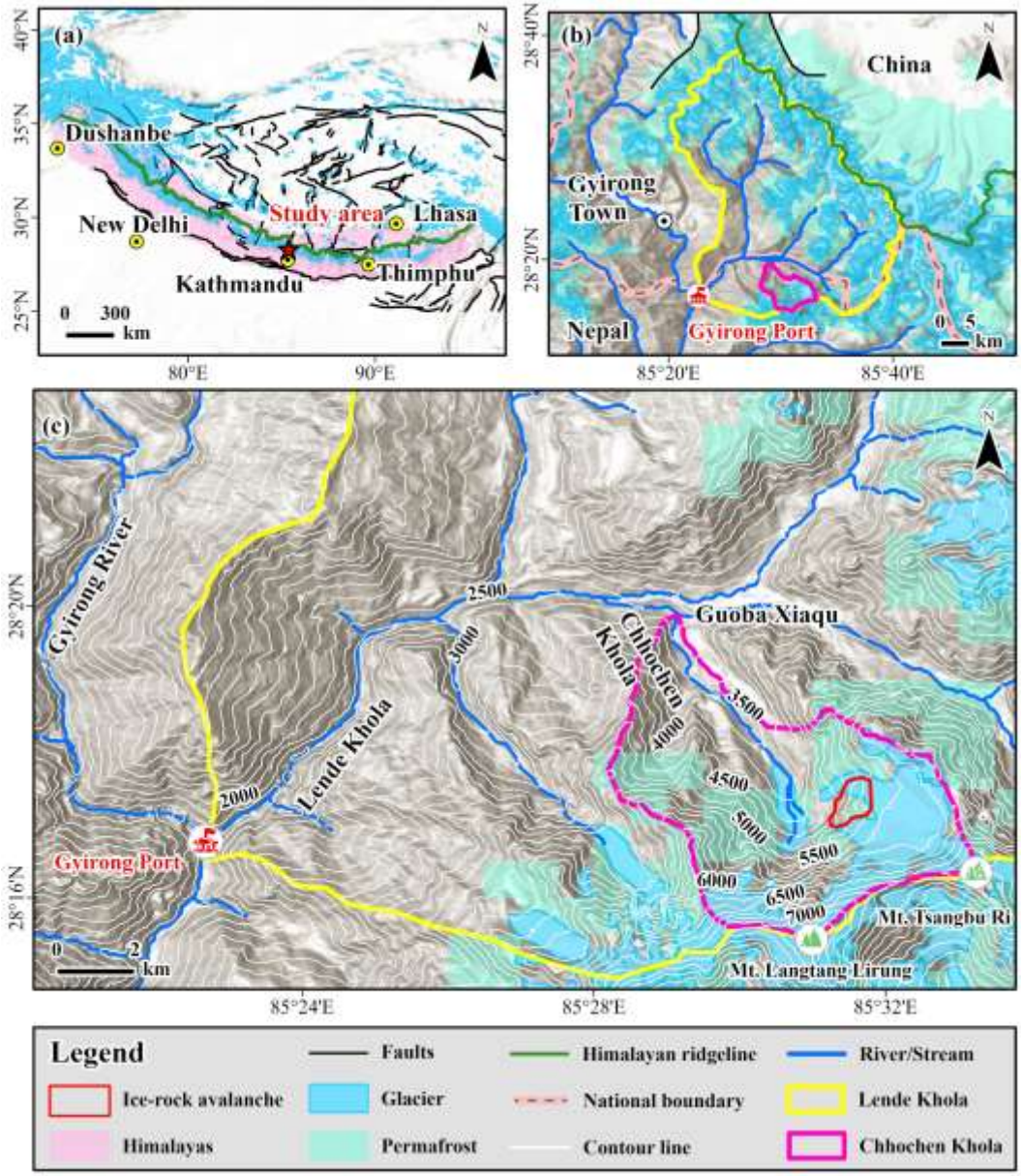


Fig. 1 Location and geographical setting of the study area. (a) Location of the study area within the Himalayas (red star), showing major nearby cities, active faults and glaciers; (b) River network, glaciers and permafrost distribution, and basin locations; (c) Topography, river network, glaciers, permafrost and ice-rock slide source area of the Chhochen Khola catchment

3 Processes of the hazard cascade

3.1 Reconstruction framework

We mapped the source, deposits, affected area, and stage boundaries using pre- and post-event optical imagery, a 30 m Copernicus DEM, an 8 m HMA DEM, field observations, and the event-induced seismic record. Images were co-registered in a common coordinate system using stable geomorphic ground-control features and a third-order polynomial transformation to reduce misalignment during change detection (*Hu et al., 2025*). We calculated normalized difference vegetation index (NDVI) from Sentinel-2 imagery acquired on 24 August 2026 and Landsat-9 imagery acquired on 26 August 2026. Pixels with NDVI below 0.19 provided an initial bare-surface classification, which we manually refined against the images to delineate the pre-event channel and event-affected area (*Li et al., 2025a*) (Fig. 2a). We used the Copernicus DEM to extract flow-path distance, elevation, and longitudinal gradient (Fig. 2b), and combined the HMA DEM with field measurements to assess representative inundated cross sections. Seismic peaks and reported arrival times constrained stage durations (Fig. 2c). The mapped affected area is approximately 15.26 km², about ten times the pre-event channel area. From the source near 5190 m a.s.l. to Gyirong Port near 1800 m a.s.l., the flow path extends approximately 22.32 km and has a mean gradient of about 14.9%. This average masks a strongly concave longitudinal profile, with gradients exceeding 55% in the upper source reach and falling below 10% in the lower debris-flow reach. The mapped deposits, slope breaks, field evidence, and multistage seismic signal recorded about 22 km from the source distinguish an ice–rock slide, an ice–rock avalanche, and a debris flow stage (Fig. 2).

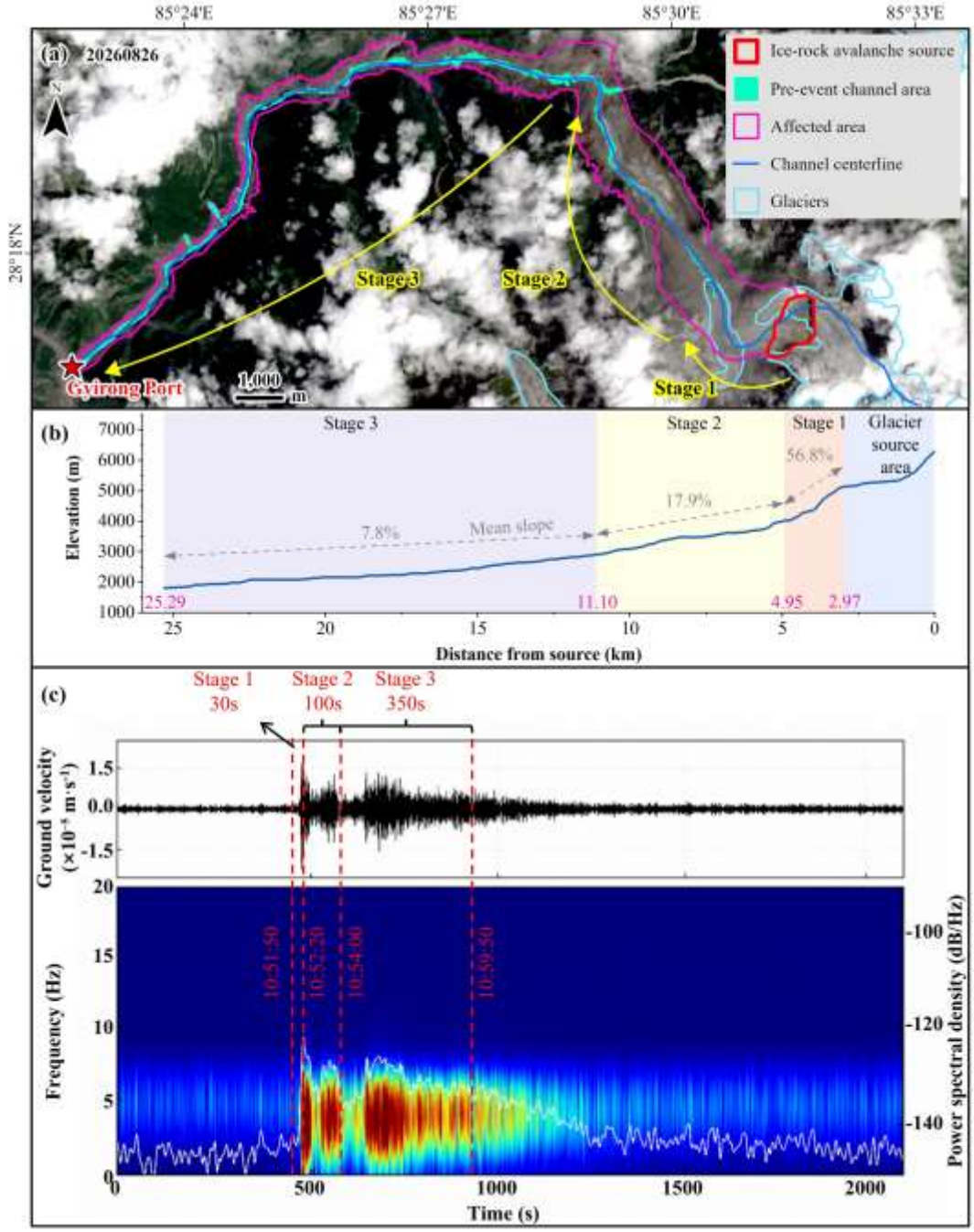


Fig. 2 Stage delineation of the cascading hazards. (a) Landsat-9 post-event image acquired on 26 August 2026 (https://vantor.com/resources/worldview-2-datasheet/); (b) Channel longitudinal profile derived from Copernicus DEM (https://panda.copernicus.eu/panda); (c) Ground-motion waveform and time-frequency spectrogram of Seismometer in Gyirong Town (Yang et al., 2026)

3.2 Stage 1 ice–rock slide

The ice–rock slide failed on the north flank of the Langtang Lirung massif at approximately 5190 m a.s.l. and reached the first impact zone near 4010 m a.s.l. after travelling 1.98 km horizontally and descending approximately 1180 m. The mean gradient of this reach was approximately 56.8% (Fig. 2). Pre- and post-event images show a distinct depletion zone and headscarp, with an approximately 620 m-wide source area (Fig. 3e). The mapped failure area is approximately 7.6×10^5 m², comprising about 2.1×10^5 m² of glacier ice and 5.5×10^5 m² of rock (Fig. 3a, c, and d). On the basis of field photographs of the exposed detachment surface, we estimate mean failure thicknesses of approximately 60 m for the ice and 165 m for the rock (Fig. 3d). The resulting component volumes are approximately 12.6 Mm³ of ice and 90.75 Mm³ of rock, giving a total of 103.35 Mm³. Ice accounts for about 12% of the source volume, and the area-weighted mean thickness is approximately 136 m.

A detached glacier tongue of approximately 0.92 km² (Fig. 3a), containing a supraglacial lake (Fig. 3c), lay below the source before failure. Detached residual ice masses can stagnate and decay, developing surface depressions, exposed ice cliffs, and supraglacial lakes (Rowan et al., 2021; Stefaniak et al., 2021). After a descent of approximately 30 s at an average speed of 66 m s^{-1}, the slide struck and entrained this tongue, enlarged the lake and impact crater, and generated a strong seismic signal equivalent to an Mw 5.1 event. The feature near 10:51:50 BJT marks the onset of movement, while the strong signal near 10:52:20 BJT corresponds to the first impact (Fig. 2c). These timing points give a Stage 1 mean velocity of approximately 66.0 m s^{-1} over the 1.98 km reach.

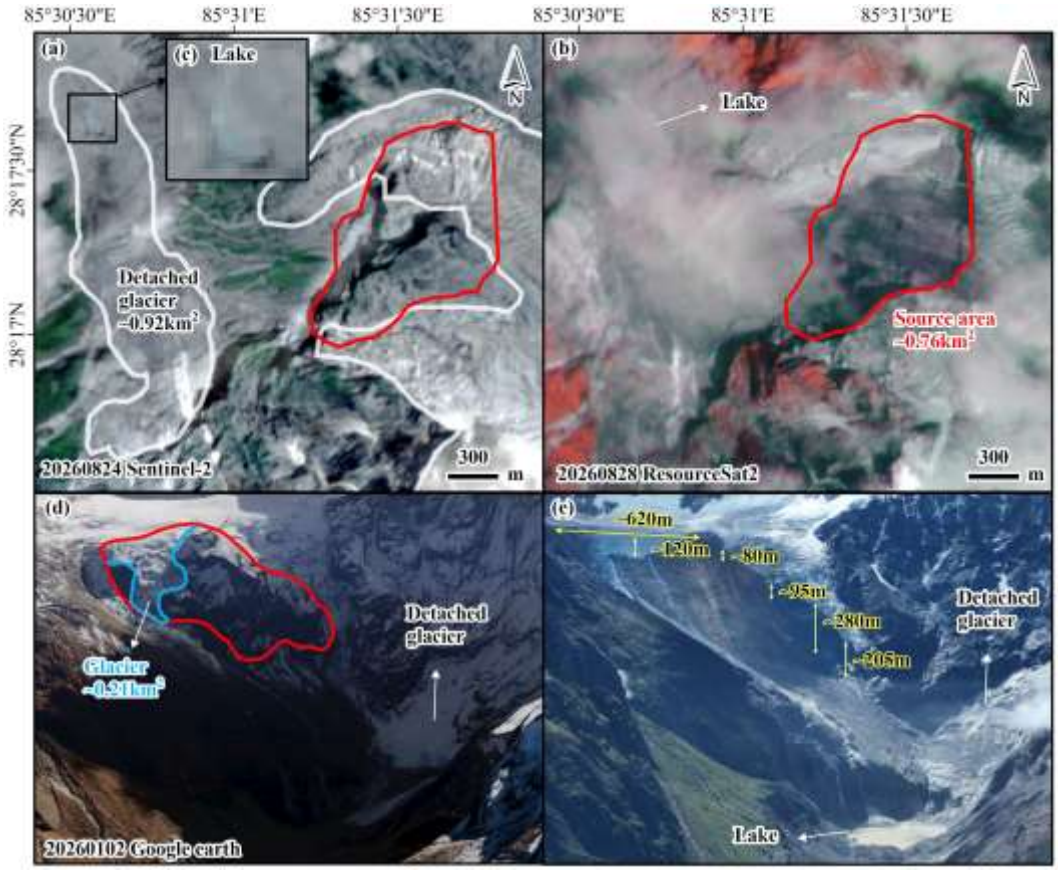


Fig. 3 Ice-rock slide stage image characteristics (a) pre-event Sentinel 2 image acquired on 24 August 2026 (https://dataspace.copernicus.eu/); (b) post-event ResourceSat 2 image acquired on 28 August 2026 (https://bhoonidhi.nrsc.gov.in/bhoonidhi/index.html#); (c) Google Earth 3D view of the source area acquired on 2 January 2026 (https://www.google.com/); (d) post-event field photograph of the source area

3.3 Stage 2: ice–rock avalanche

After the first high-energy impact, fragmented ice and rock mixed with entrained glacier ice and water from the supraglacial lake, developing into a large ice–rock avalanche along Chhochen Khola. A second pronounced seismic peak near 10:54:00 BJT marks arrival at the Guoba Xiaqu confluence and collision with the opposite hillslope, forming a second high-energy impact zone. The avalanche travelled approximately 6.15 km, descended about 1100 m, and crossed this reach in approximately 100 s (Fig. 2). Its mean gradient was about 17.9% and its timing-derived mean horizontal velocity was approximately 61.5 m s^{-1}. The slope break near 4010 m and the trough-shaped valley coincided with impact,

fragmentation, and substantial deposition. Pre- and post-event images and field observations show that much of the material was stored in Chhochen Khola (Fig. 4). We mapped trough-valley deposits over approximately 1.4 Mm^2 and other deposits over approximately 2.5 Mm^2. Representative mean thicknesses were estimated at about 30 m and 10 m, respectively (Fig. 4b-d), using field photographs, the DEM, and valley cross-sectional geometry. The total deposit estimate is approximately 67 Mm^3. The remaining approximately 36 Mm^3 entered the Lende Khola and supplied the downstream debris flow.

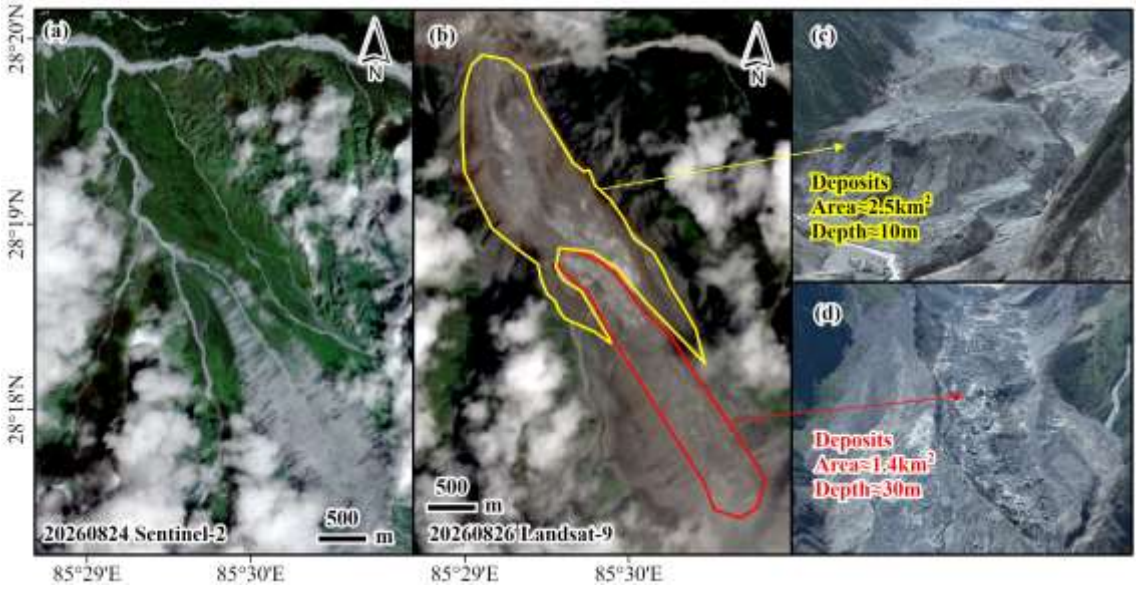


Fig. 4 Ice–rock avalanche stage: (a) pre-event Sentinel 2 image acquired on 24 August 2026; (b) post-event Landsat-9 image acquired on 26 August 2026; (c–d) Chhochen Khola deposits.

3.4 Stage 3: debris flow

At the confluence of the Chhochen Khola and Guoba Xiaqu, the moving mass turned westward into the Lende Khola and transformed into a debris flow. The reach from the confluence to Gyirong Port was approximately 14.19 km long, with a mean gradient of about 7.8%. The flow reached Gyirong Port at approximately 10:59:50 BJT, after about 350 s (Fig. 2), giving a timing-derived Stage 3 mean velocity of approximately 40.5 m s^{-1}. The affected corridor remained largely valley-confined but widened locally at bends and broader valley sections (Fig. 5b).

At the bedrock-gorge near the Gyirong Port, field-surveyed mud marks indicated a maximum flow depth of approximately 50 m and a maximum inundated cross-sectional area of 6790 m² (Fig. 5d-f). We calculated the mean flow velocity using Manning's equation, which is expressed as:

$$v = \frac{1}{n} * R^{2/3} * J^{0.5} \quad (1)$$

where v is the mean flow velocity (m/s), J is the channel bed slope (0.063), and n is Manning's roughness coefficient (0.065). The hydraulic radius (R) can be derived by $R=A/P$ (m), where A is the cross-sectional area (6790 m²), P is the wetted perimeter (236.18 m). The reconstructed cross-sectional mean velocity was 36.2 m s^{-1} and the peak discharge was 2.46 × 10^5 m^3 s^{-1}. Pre- and post-event images show pronounced changes in the channel and surrounding terrain: nearly all buildings and other surface features in the affected port area were removed (Fig. 5c-d). After entering the Gyirong River, material continued downstream while a substantial sediment volume propagated nearly 3 km upstream into Chinese territory. The channel widened markedly and the riverside road was destroyed (Fig. 5b).

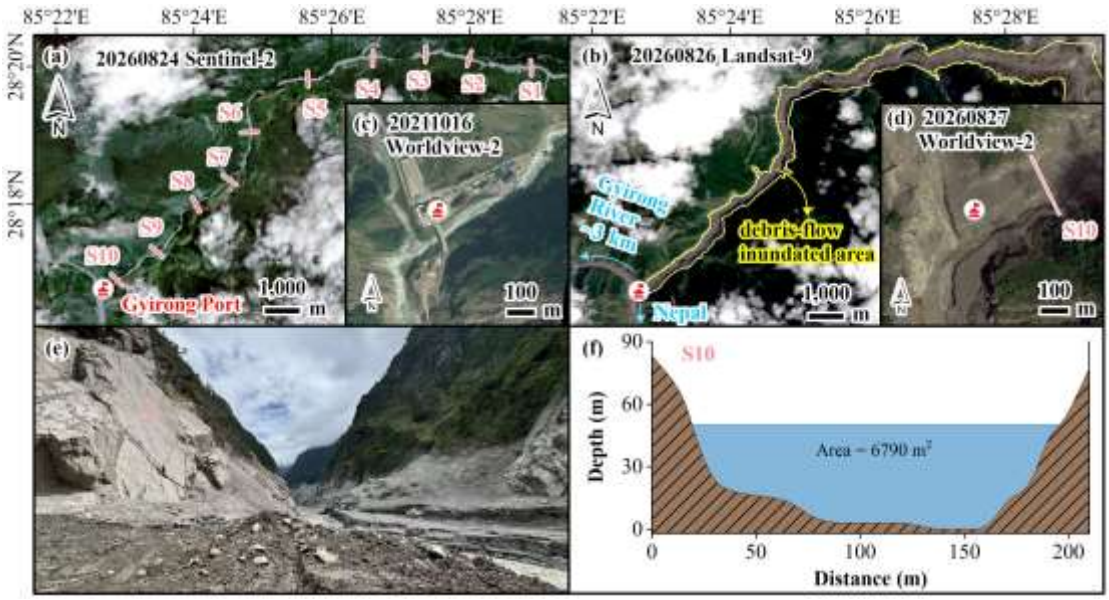


Fig. 5 Downstream debris-flow satge: (a) pre-event Sentinel 2 image acquired on 24 August 2026 and control sections; (b) post-event Landsat-9 image acquired on 26 August 2026; (c–d) WorldView-2 images of Gyirong Port acquired on 16 October 2021 and 27 August 2026

(https://browser.moregeo.it/external/vantor-opendata.s3.amazonaws.com/events/Nepal-Flooding-Aug-2026/collection.json); (e-f) Field photograph and schematic diagram near cross-section S10.

Across the ice–rock slide, ice–rock avalanche, and debris-flow stages, timing-derived mean velocities decreased from approximately 66.0 to 61.5 and 40.5 m s^{-1} as mean channel gradients fell from approximately 56.8% to 17.9% and 7.8% (Fig. 6). The approximately 40% decrease from the first to the third stage coincided with the downslope decrease in gradient and extensive intermediate storage. These values describe travel between seismic or arrival-time constraints. The complete 22.32 km source-to-port transit took approximately 8 min, yielding an overall mean velocity of about 46.5 m s^{-1}. The stage timing and slope breaks provide constraints on the transformation and downstream arrival of this cascade.

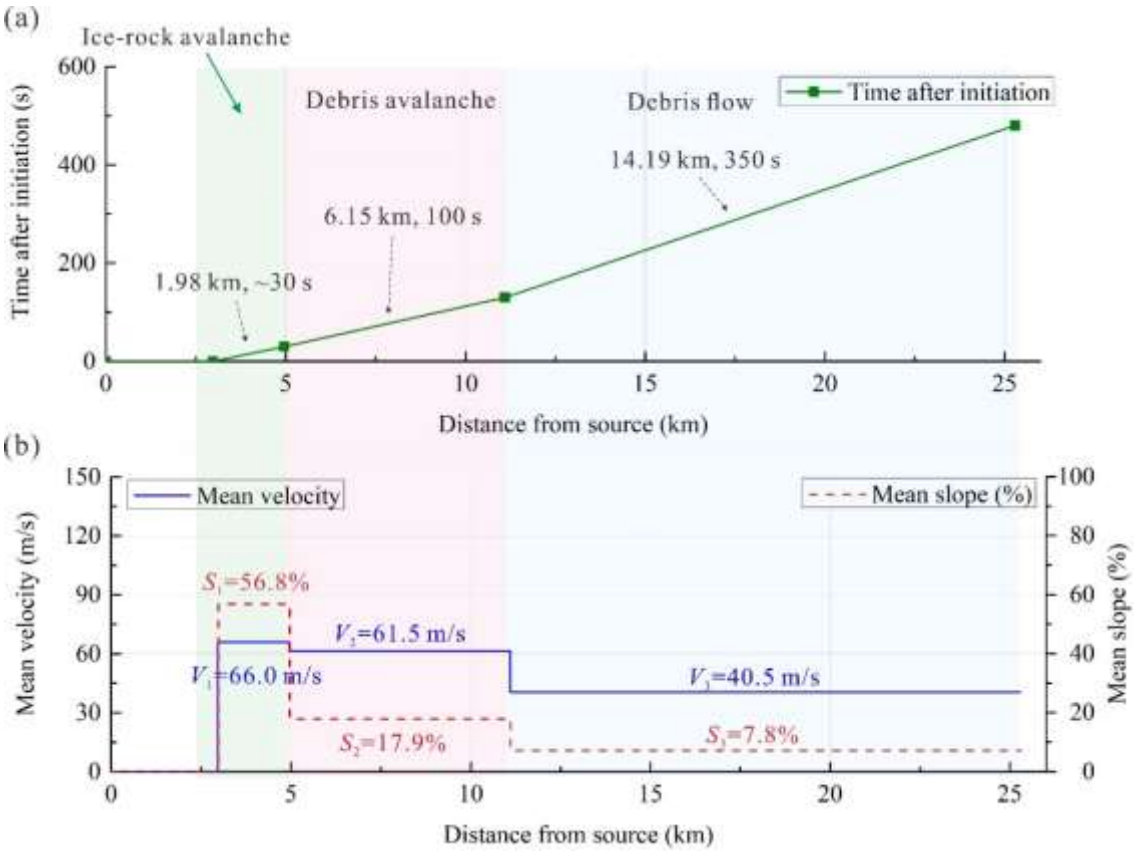


Fig. 6 Kinematic evolution of the 26 August 2026 cascade. (a) Distance-time reconstruction for the three stages. (b) Timing-derived stage-averaged velocities and corresponding mean channel gradients.

4 Numerical Computation of the Ice–Rock Slide and Debris-Flow Routing

The disaster chain was modelled in two of its three stages. The debris-avalanche stage (Stage 2) was not simulated because field investigations revealed extensive deposits along this reach, indicating that a substantial fraction of the moving mass was deposited before reaching the distal channel. A dynamic back-analysis of this stage would therefore require detailed constraints on entrainment and deposition that are not currently available. Accordingly, we focus on (i) a rigid-slider model with Voellmy turbulent resistance for the ice-rock slide (Stage 1), and (ii) a one-dimensional hydraulic-routing model for the debris-flow stage (Stage 3), which was responsible for the majority of casualties.

4.1 Stage 1: ice–rock slide

We approximated the initial motion of the detached mass with a one-dimensional rigid-slider model incorporating gravity, Coulomb friction, and Voellmy turbulent resistance (*Hungr and McDougall, 2009*). The governing equations are:

$$\frac{\mathrm{d}u}{dt} = g\sin\theta - \mu g\cos\theta - \frac{u^2 g}{\xi h} \tag{2}$$

$$\frac{ds}{dt} = u \tag{3}$$

where u is the sliding velocity, s is the travel distance along the slope, θ is the local slope angle, μ is the Coulomb friction coefficient, ξ is the turbulence coefficient (m/s²), and h is a reference flow depth. For the ice-rock slide, the μ and ξ were set to 0.05 and 1200 m s^{-2}, respectively. Both values fall within the typical ranges reported for large rock–ice avalanches (μ = 0.05–0.2; ξ = 1000–4000 m s^{-2}). The relatively low μ reflects the moderate basal resistance of a very thick (~136 m) fragmented ice–rock mass, while the moderate ξ accounts for the suppressed turbulence intensity per unit volume in such a thick flow. The velocity–time curve computed from the rigid-slider model (Fig. 7) shows a rapid acceleration followed by saturation near the steady-state value, consistent with the

behaviour expected from the Voellmy turbulent resistance term. Using the Stage 1 mean slope and a representative flow depth of ~136 m, the mean velocity computed from Eqs. (2-3) is 64.2 m s^{-1} over a travel distance of 1.98 km. This is close to the seismic-inferred mean velocity of 66.0 m s^{-1}, with a difference of 1.8 m s^{-1} (approximately 2.7%). The terminal velocity at the end of Stage 1 is 129.2 m s^{-1}, reflecting the limited travel distance over which the mass accelerated before entering the transitional ice-rock avalanche reach.

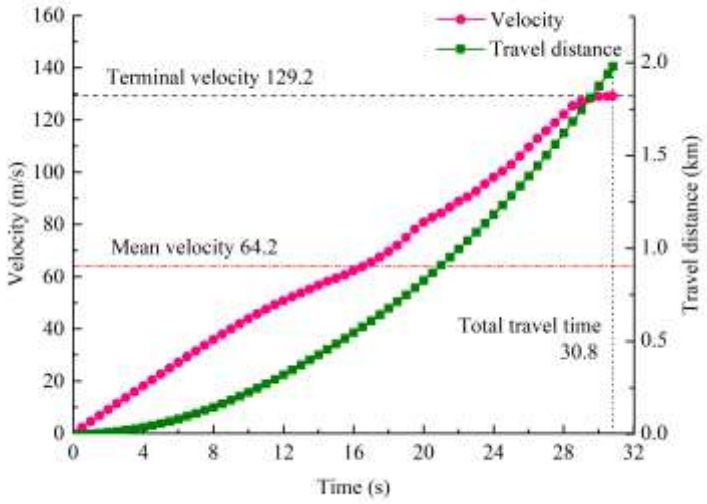


Fig. 7 Velocity–time curve and travel distance for the Stage 1 ice–rock slide computed with the rigid-slider model and Voellmy turbulent resistance. The calculation uses a 1.98 km computational distance.

4.2 Stage 3: debris-flow routing

The Stage 3 model is used as a simplified one-dimensional hydraulic-routing model rather than a complete process-based debris-flow model. Its purpose is to characterize the equivalent longitudinal evolution of the debris flood, particularly peak-discharge attenuation, travel time, and hydrograph broadening. We routed the approximately 36 Mm^3 of material entering the Lende Khola with a one-dimensional, depth-averaged Saint-Venant shallow-water model. The equations were written in area–discharge form (*Hodges, 2019*):

$$\frac{\partial A}{\partial t}+\frac{\partial Q}{\partial x}=0 \tag{4}$$

$$\frac{\partial Q}{\partial t}+\frac{\partial}{\partial x}\left(\beta\frac{Q^2}{A}+gI_1\right)=gI_2-gA\frac{\partial z_b}{\partial x}-gAS_f \tag{5}$$

where A is the wetted cross-sectional area, Q is discharge, x is the streamwise coordinate, t is time, g is gravitational acceleration, z_b is bed elevation, S_f is the friction slope, and β is the momentum-flux correction coefficient. In the present implementation β is set to 1. I_1 and I_2 are integrated hydrostatic pressure forces across the channel and along the channel, respectively. Manning resistance is expressed as:

$$S_f=\frac{n^2Q|Q|}{A^2R^{4/3}} \tag{6}$$

Each computational cross section is represented by a symmetric trapezoid with bottom width b, side slope m expressed as horizontal distance per unit vertical distance on each bank (larger m values representing gentler side slopes), and flow depth h. The principal geometric relations are:

$$A=bh+mh^2 \tag{7}$$

$$I_1=\frac{1}{2}bh^2+\frac{1}{3}mh^3 \tag{8}$$

$$I_2=\frac{1}{2}h^2\frac{db}{dx}+\frac{1}{3}h^3\frac{dm}{dx} \tag{9}$$

The equations were discretized with a second-order finite-volume method. The left and right states at each cell interface were obtained using MUSCL reconstruction (*van Leer, 1979*), with a minmod slope limiter employed to satisfy the total-variation-diminishing condition and suppress spurious numerical oscillations (*Harten, 1997*). The intercell numerical fluxes were evaluated using the HLL approximate Riemann solver, and temporal integration was performed using a second-order strong-stability-preserving Runge–Kutta scheme (*Shu and Osher, 1988*). The time-step size was automatically adjusted according to the flow velocity, wave speed, and grid spacing to maintain numerical stability.

The Stage 3 domain comprised 10 control cross sections (S1–S10) derived from Copernicus topography and remote-sensing interpretation (Fig. 5a; Table 1). Each cross section is idealized as a symmetric trapezoid defined by its bottom width b (28.9-108.0 m), side-slope coefficient m (0.86-2.69), and bed elevation z_b (1834-2886 m). The Manning coefficient n is uniformly assigned a value of 0.065 for S1-S10 (Table 1). The geometric parameters are linearly interpolated by channel chainage onto a nominal 30 m computational grid. At the upstream boundary, flow velocity is fixed at 50 m s^{-1}. The flow-depth boundary condition is prescribed as a monotonically decreasing function, decreasing from 130 m to 0 over a total duration of approximately 33.39 s. For the trapezoidal geometry of S1, the prescribed peak wetted area is 5.8×10^4 m^2 and the corresponding prescribed boundary discharge is approximately 2.914×10^6 m^3 s^{-1}. The assumed event volume is approximately 3.60×10^7 m^3, and the total simulation time is 500 s.

Table 1 Parameters of the selected control cross sections.

Location	b (m)	m	z_b (m)	Manning coefficient n
S1	98.8	2.69	2886	0.065
S2	92.0	2.28	2687	
S3	36.3	2.69	2608	
S4	35.9	1.73	2421	
S5	28.9	2.23	2286	
S6	80.6	1.72	2169	
S7	108.0	1.79	2079	
S8	63.3	1.65	2017	
S9	63.9	1.75	1911	
S10	92.7	0.863	1834	

The modeled hydrographs show progressive peak attenuation, delayed peak arrival, and temporal broadening along the channel (Fig. 8a, b). The prescribed inlet discharge at S1 peaks at 2.914×10^6 m^3 s^{-1} at t = 0 s. At S10, immediately upstream of Gyirong Port, discharge peaks at 2.408×10^5 m^3 s^{-1} at t = 270 s, representing an attenuation of 91.74%. The corresponding peak-to-peak travel time is 270 s, giving a mean peak-propagation

velocity of 48.97 m s^{-1}. The maximum velocity, flow depth, and wetted area at S10 are 40.16 m s^{-1}, 48.41 m, and 6,509 m^2, respectively, which were close to our field measurements and estimates (Fig.5 and Table 2). To quantify downstream hydrograph broadening, we used the full width at half maximum (FWHM, denoted W50), which is calculated as:

$$W_{50} = {}^{t}Q/Q_{p}=0.5,\text{fall} - {}^{t}Q/Q_{p}=0.5,\text{rise} \tag{10}$$

Hydrograph broadening was substantial, with *W50* increasing from 10.96 s at the prescribed S1 boundary to 111.47 s at S10, an absolute increase of 100.51 s and a 10.17-fold final-to-initial ratio. Because the inlet hydrograph begins at its peak at t = 0 s, the S1 value represents the one-sided duration from the prescribed peak to the falling half-peak crossing (Fig. 8 and Table 2). This behavior reflects longitudinal redistribution of the debris-flow volume, expressed by progressive peak attenuation and extension of the recession limb during downstream propagation.

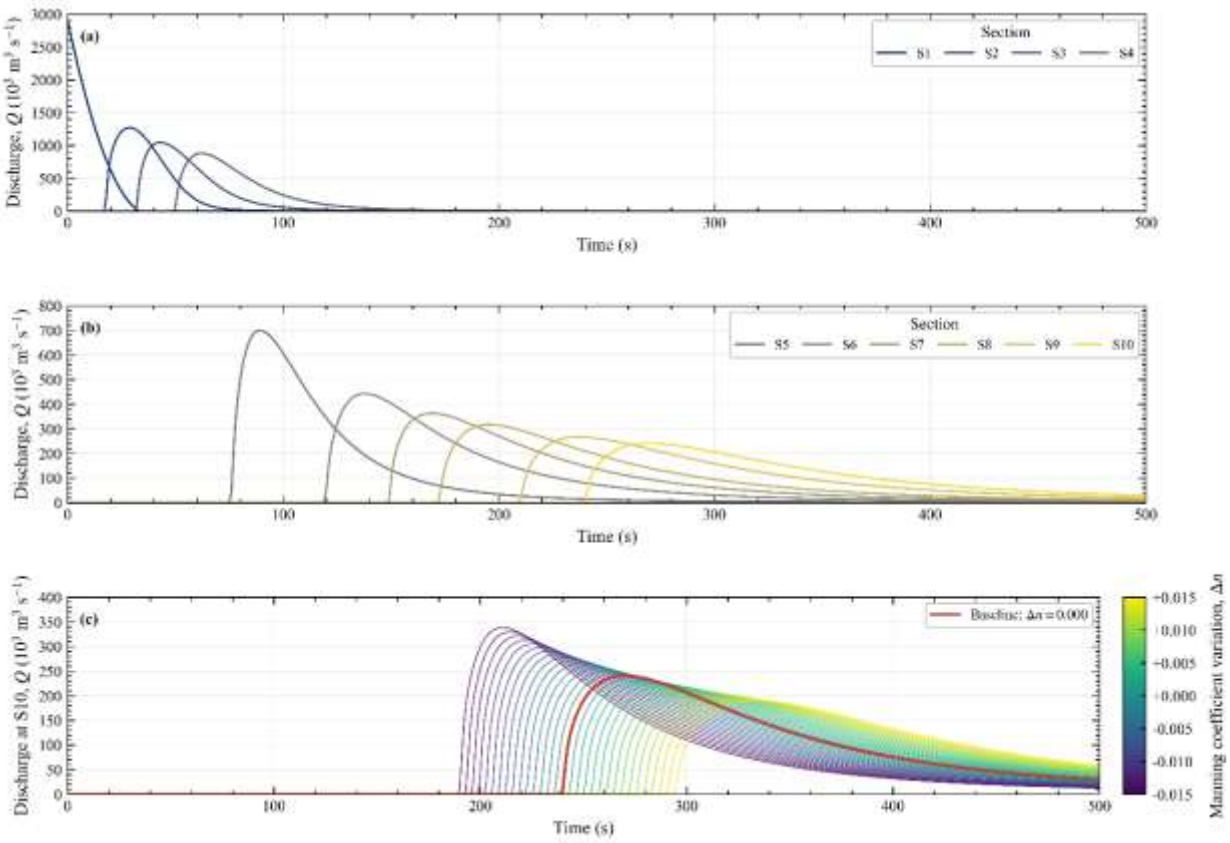


Fig. 8 Simulated discharge hydrographs at the 10 control cross sections. (a) S1–S4. (b) S5–S10. (c) S10 hydrographs for the Manning-coefficient sensitivity scenarios.

Table 2 Peak hydraulic characteristics at the 10 control cross sections.

Location	Peak discharge ($m^3 s^{-1}$)	Time to peak discharge (s)	Peak velocity ($m s^{-1}$)	Peak flow depth (m)	W_{50} (s)
S1	2913759	0	50.00	130.00	10.96
S2	1271982	29	75.76	73.41	27.36
S3	1029057	43	67.04	72.80	30.76
S4	881044	62	69.16	79.14	34.95
S5	698891	89	59.83	69.23	41.32
S6	440973	138	48.75	57.46	62.01
S7	362665	169	44.88	45.78	75.35
S8	315194	196	43.17	54.73	85.55
S9	266670	237	40.67	47.72	101.32
S10	240770	270	40.16	48.41	111.47

Note: The inlet hydrograph begins at its peak at t = 0 s; W50 at S1 is reported as the one-sided duration to the falling half-peak crossing.

A grid-resolution sensitivity analysis was conducted using nominal spatial resolutions of 15, 30, and 60 m, while all other model parameters and boundary conditions were kept unchanged. At the downstream control section (S10), the simulated peak discharge ranged from 240,560 to 240,812 m^3/s, the peak-arrival time remained unchanged at 270 s, and the maximum flow velocity ranged from 39.89 to 40.29 m/s. Relative to the 30 m baseline, the differences in peak discharge, maximum flow depth, and *W50* were within 0.09%, 0.05%, and 0.03%, respectively, while the difference in maximum velocity remained below 0.7%. These small variations indicate that the principal downstream results are only weakly affected by grid resolution and that the adopted 30 m grid provides an adequate balance between numerical accuracy and computational efficiency. A sensitivity analysis was conducted by uniformly varying the Manning coefficient from 0.050 to 0.080 for S1–S10, while all other model parameters and boundary conditions were kept unchanged. A total of 31 scenarios were simulated at an increment of 0.001. At S10, the peak discharge decreased from 338,550 to 189,802 m^3/s (Fig. 8c), the peak-arrival time increased from 211 to 334 s, and the maximum flow velocity decreased from 54.24 to 32.05 m/s. These results demonstrate that the assumed flow resistance has a substantial influence on the simulated downstream propagation and peak-flow characteristics.

The simulated mean peak-propagation velocity from S1 to S10 is 48.97 $m\ s^{-1}$, which is 8.47 $m\ s^{-1}$ (approximately 17.3%) higher than the observational estimate of 40.5 $m\ s^{-1}$. This result indicates that the model reproduces the rapid downstream propagation of the debris flood but moderately overestimates its overall propagation rate. The discrepancy may primarily result from the one-dimensional representation of the channel, which cannot explicitly resolve channel bends, lateral momentum redistribution, flow superelevation, lateral spreading, tributary backwater effects, or the additional energy dissipation associated with these two-dimensional processes. The use of simplified cross-sectional geometry and an equivalent Manning coefficient may also underestimate the effective flow

resistance, while uncertainties in the reconstructed pre-event topography and the observationally estimated propagation velocity could further contribute to the difference between the simulated and observed values.

5 Potential contributors to ice-rock failure

5.1 Immediate meteorological and seismic conditions

For the pre-event conditions, we converted half-hourly GPM IMERG rainfall to Beijing time (UTC+8) and aggregated it into daily totals. ERA5-Land daily mean air temperature was examined for the three months preceding failure. We examined GPM IMERG rainfall and the USGS earthquake catalogue (https://earthquake.usgs.gov/earthquakes/) before the event. Rainfall was generally low in the preceding months, and no extreme rain event or temperature anomaly immediately preceded failure (Fig.8a). No earthquake above M 4.5 occurred within 50 km of the source in the preceding 72 h. The Mw 5.1-equivalent signal was generated during the cascade at the ice-rock impact, after the slide had begun.

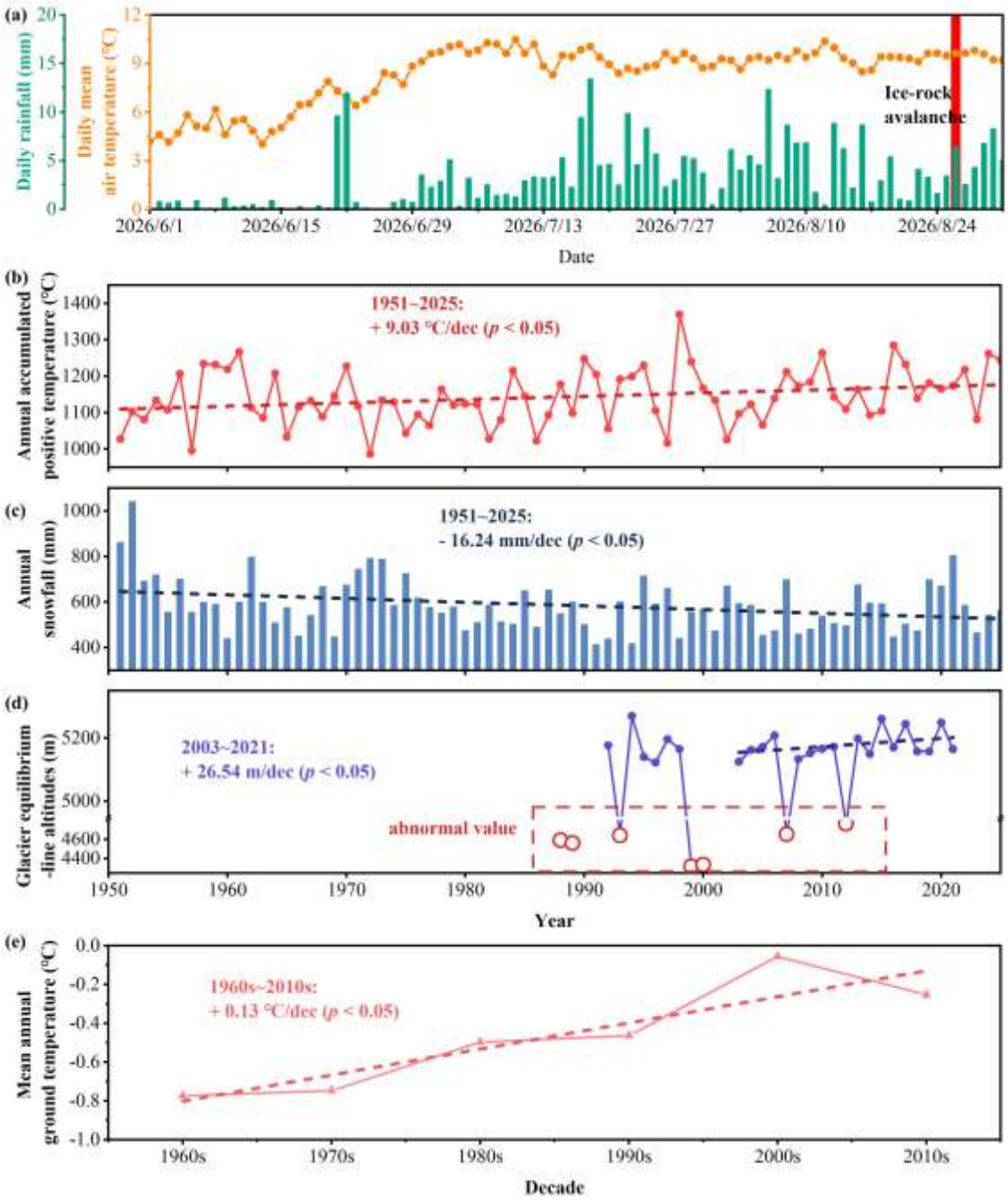


Fig. 8 Meteorological and cryospheric context: (a) ERA-5 daily temperature and GPM rainfall preceding 3 months (Acquired online via Google Earth Engine); (b) annual positive accumulated temperature from 1951-2025 (Acquired online via Google Earth Engine); (c) annual snowfall from 1951-2025 (Acquired online via Google Earth Engine); (d) glacier equilibrium-line altitude from 1985-2021(Loibl et al., 2025); (e) mean annual ground temperature from 1960s to 2010s (https://cstr.cn/18406.11.Cryos.tpdc.272790). Dashed lines show trends; open circles show excluded outliers

5.2 Climatic and cryospheric context

For long-term context, we calculated annual positive accumulated temperature and snowfall for 1951–2025, as well as the mean annual ground temperature of the source area from 1960s to 2010s. The remotely sensed glacier snowline dataset spans 1985–2021; we used the maximum snowline altitude in each melt season as a proxy for glacier equilibrium-line altitude (ELA) (*Loibl et al., 2025*). The target glacier is represented by RGI60-15.04172 and RGI60-15.04173 in RGI 6.0 and by the merged RGI2000-v7.0-G-15-05732 in RGI 7.0; we averaged the annual maxima of the two older units and excluded marked interannual outliers when estimating the 2003–2021 trend. Annual positive accumulated temperature increased by 9.03 °C decade^{-1}, annual snowfall decreased by 16.24 mm decade^{-1}, ELA rose by 26.54 m decade^{-1}, and mean annual ground temperature increased by approximately 0.13 °C decade^{-1} (all $p < 0.05$) (Fig.8b-e). Mean air temperature in the three months before failure was 8.11 °C, slightly below the 1951–2025 climatological mean of 8.44 °C. These records show cumulative warming and cryospheric change without an immediate temperature extreme.

Regional studies show that glacier retreat, snowline rise, freeze–thaw action, and warming of ice-filled fractures can alter the thermal and mechanical conditions of steep rock slopes (*Deline et al., 2015*; *Gruber and Haeberli, 2007*; *Krautblatter et al., 2012*). In the adjacent Langtang catchment, glacier area has declined by approximately 41.5% since the Little Ice Age, with losses accelerating after 2000; rising equilibrium-line altitudes increasingly intersect steep, narrow, crevassed glacier sections and favour fragmentation and disconnection (*Silwal et al., 2026*). Glacier-related mass movements along the Gyirong corridor are associated with glacierized terrain, temperature, precipitation, and steep topography (*Qiu et al., 2022*). Records from the eastern Himalaya also connect warming with increased periglacial debris-flow activity (*Li et al., 2025b*). Meltwater penetration into joints, repeated freezing and thawing, changing glacier support, and thawing of ice-filled fractures provide plausible pathways for progressive weakening, although these processes have not been directly measured at the Chhochen Khola source.

5.3 Geological context

The Langtang–Gyirong sector lies within the actively deforming central Himalaya, underlain by the Main Himalayan Thrust and accompanied by major extensional structures in southern Tibet. The 2015 Mw 7.8 Gorkha earthquake triggered a catastrophic ice, snow, and rock avalanche in Langtang Valley and produced widespread landslides, rockfalls, collapses, and ground fissures across the border region (*Fujita et al., 2017*; *Gnyawali et al., 2020*; *Wu et al., 2019*). Reported shaking intensity reached VIII–IX in parts of this region (*Wu et al., 2019*). Historical shaking and tectonic deformation may have left inherited fracture networks and damaged rock bridges; the absence of a nearby earthquake immediately before the 2026 event distinguishes such preconditioning from direct coseismic triggering.

Together, the records favour a compound origin involving long-term climatic change and adverse geological conditions. Warming, glacier retreat, reduced snowfall, freeze–thaw cycling, inherited fractures, and changing glacier support could have progressively weakened the source slope. However, the available observations do not separate the contribution of each process to the final failure.

6 Discussion and limitations

6.1 Mass redistribution and stage transformation

The volume inventory shows how the cascade redistributed mass before reaching the main river. The approximately 103 Mm^3 source comprises about 12.6 Mm^3 of glacier ice and 90.4 Mm^3 of rock. The adopted bulk estimate is about 67 Mm^3 retained within Chhochen Khola and a residual of approximately 36 Mm^3 supplied to the Lende Khola. This balance links the source to the downstream boundary condition: only about one third of the estimated source volume continued into the debris-flow reach. However, areas and representative thicknesses are approximate and do not resolve spatially variable erosion or deposition.

The three process reaches occupy different parts of a strongly concave longitudinal profile. Timing-derived mean velocities decrease from 66.0 m s^{-1} in the ice–rock slide to 61.5 m s^{-1} in the ice–rock avalanche and 40.5 m s^{-1} in the debris flow, while their mean gradients fall from 58.8% to 15.6% and 7.1%. The decrease is consistent with reduced gravitational driving force and substantial storage along Chhochen Khola, although a stage-averaged velocity alone cannot isolate the effects of slope, impact, friction, or material composition.

The available observations identify several possible contributions to the flow transformation without requiring any one source of water to dominate. The failed mass itself contained an estimated 12.6 Mm^3 of ice. The detached glacier tongue and its supraglacial lake supplied additional ice and water at the first impact, and the moving material could subsequently interact with channel water and valley-floor sediment. Concurrently, much of the coarse bulk accumulated in Chhochen Khola. The coexistence of water acquisition and deposition provides a coherent explanation for a smaller but still destructive mobile fraction entering the Lende Khola. No direct water budget or deposit-grain-size transect is available to apportion these contributions, so the stage names describe observed reach-scale behaviour rather than a measured sequence of rheological states.

6.2 Comparison with other studies

Recent studies provide broadly consistent evidence for a mixed ice–rock source, but differ in how they define and interpret it. *Guo et al. (2026)* mapped a 0.72 km^2 mixed ice–rock source. They found sustained glacier motion before the failure and interpreted channel erosion and entrainment as important to the flow's growth. *Zhao et al. (2026*) mapped a similar source area of about 0.75 km^2 and identified bedrock fractures, glacier retreat and possible permafrost degradation as factors that weakened the slope. Neither study established an immediate trigger. These observations accord with our interpretation of a rock-dominated failure conditioned by geological structure and cryospheric change. Our deposit mapping adds an important qualification to their emphasis on erosion: much of

the material was stored in the valley before the remaining flow reached the main river. *Yang et al. (2026)* instead described the source mainly as an ice avalanche, estimating an area of 0.55 km² and a volume of 20–40 Mm³. The smaller estimate may follow from a different mapped boundary and assumed thickness. The studies also locate the change in flow type differently. *Guo et al. (2026)* describe progressive erosion and entrainment, and place the transition to debris flow near 4,000 m a.s.l., where the moving material mixed with water and sediment. *Zhao et al. (2026)* classify the upper reach as an ice–rock avalanche and describe its transformation into a debris flow after it entered the main river near 2,930 m a.s.l. *Yang et al. (2026)* describe a progression from ice avalanche through debris avalanche to debris flow, using broad zones rather than a single transition point. Our reconstruction distinguishes an initial ice–rock slide from the avalanche formed after glacier impact, and places the subsequent debris-flow stage at the main-river confluence.

The Chhochen Khola event shares the high-elevation rock–ice source, rapid initial motion, and damaging downstream transformation documented at Chamoli in 2021 (*Shugar et al., 2021*; *Zhang et al., 2023*). Both disasters involved initial velocities of the order of 50–100 m s^{-1}, developed a debris flow after several kilometres of travel, and affected infrastructure far from the source. At Chhochen Khola, slope breaks near 4010 and 2910 m a.s.l., the timed glacier impact, and the large mapped depositional reach make the separation between the ice–rock slide, ice–rock avalanche, and debris flow particularly clear. These transitions are more spatially discrete than the transition described for Chamoli, plausibly reflecting differences in channel geometry, sediment availability, and water interaction (Table 3).

Table 3 Comparison of the 2026 Chhochen Khola event and the 2021 Chamoli disaster.

Parameter	2026 Chhochen Khola event	2021 Chamoli event
Source elevation (m a.s.l.)	~5190	~5500
Initial volume (m^3)	~1.03×10^8	~2.7×10^7
Stage 1 velocity (m s^{-1})	~66.0	~60

Total runout (km)	~22.32	~26
Stage 2 length (km)	6.15	~8
Stage 3 present?	Yes, well-defined	Less distinct
Trigger	Compound climatic and geological preconditioning	Climatic (permafrost/glacial)

The 2016 Aru glacier detachments in western Tibet provide a high-elevation cryospheric comparison, but involved the collapse of glacier tongues after surge-like instability and reduced basal resistance (*Kääb et al., 2018*). In contrast, the Chhochen Khola source was predominantly rock, and the moving mass subsequently struck a detached glacier before extensive deposition in the valley. The 2015 Langtang disaster was triggered by the Gorkha earthquake. Its primary avalanche was interpreted as predominantly snow, followed by rockfalls; a separate numerical reconstruction examined interacting snow–ice and rock-debris flows from different source areas (*Fujita et al., 2017*; *Gnyawali et al., 2020*). These cases illustrate distinct source mechanisms and material-transfer histories. The Chamoli event also shows why flow transformation requires evidence from more than one part of the route. *Wang et al. (2024)* used deposit observations and discharge reconstructed from flow superelevation to infer substantial channel erosion and river-water incorporation. They argued that the contribution of rapid source-ice melting to mobility may have been overestimated. At Chhochen Khola, the glacier impact and supraglacial lake indicate plausible water inputs, while valley deposition and changing channel confinement are also documented. The comparison supports assessing impact, entrainment, storage, and routing stage by stage.

6.3 Implications for hazard assessment

Hazard assessment in glacierized Himalayan valleys should follow the moving mass from the unstable source to the exposed downstream corridor. For this event, source mapping alone would have overestimated the material entering the Lende Khola because approximately 67 Mm^3 was stored in Chhochen Khola. Conversely, the remaining

approximately 36 Mm^3 was sufficient to generate an exceptionally large downstream peak. Maps of potential impact points, detached glacier, lakes, depositional troughs, confluences, and confined gorges are therefore needed alongside source inventories. The three stages carry different useful metrics: source and impact timing for the slide, travel and storage for the avalanche, and discharge, depth, arrival time, and attenuation for the debris flow.

The routing results show why downstream intensity cannot be inferred from transfer volume alone. With the same 36 Mm^3 event volume, the modeled hydrograph attenuates from a prescribed S1 peak of 2.914×10^6 m^3 s^{-1} to about 2.408×10^5 m^3 s^{-1} at S10, while W50 expands from 10.96 to 111.47 s. Thus the modelled peak becomes lower and broader as it travels through the Lende Khola. The separate bedrock-gorge estimate of 2.46×10^5 m^3 s^{-1} near Gyirong Port supports the order of magnitude of the distal peak, whereas the Manning sensitivity range demonstrates how strongly the precise value depends on effective resistance. Precomputed scenarios should therefore cover a range of resistance and inlet conditions when used for planning.

The roughly 8 min source-to-port transit leaves limited time between detection and arrival at Gyirong Port. The first large impact occurred about 30 s after initial movement, and the main-river confluence was reached roughly 100 s later. A warning system based on this chronology would need rapid detection, automatic estimation of affected downstream reaches, and established cross-border communication routes. Seismic signals can supply early event timing, while mapped stage boundaries and precomputed routing scenarios can indicate where a moving flow is likely to arrive. The practical warning interval depends on when an event is detected and how quickly alerts reach exposed communities; the event duration itself is not a guaranteed lead time.

6.4 Constraints and future work

The source and deposit volumes are constrained mainly by mapped planform areas, field photographs of the detachment surface, representative thicknesses, and valley geometry.

Higher-resolution before-and-after DEMs, seismic inversion, and systematic field thickness surveys would refine the estimates and their spatial mass balance. The 12.6 Mm^3 source-ice estimate describes only the initial failed mass; ice entrained from the detached glacier remains unmeasured. Seismic peaks and arrival reports provide discrete stage timing rather than continuous trajectories, so reported reach means and the modeled local Stage 1 maximum describe different aspects of motion.

The Saint-Venant model represents longitudinal routing through idealized trapezoidal sections with a prescribed inlet hydrograph and uniform effective Manning resistance. Its grid test shows numerical stability over 15–60 m spacings, while the resistance scenarios give substantial variation in S10 peak discharge, arrival time, and velocity. The baseline S10 peak agrees in order of magnitude with the independent bedrock-gorge estimate, but the modeled S1–S10 peak propagation is approximately 22% faster than the observational estimate. Bends, lateral momentum redistribution, flow superelevation, bank impact, tributary backwater, sediment entrainment and deposition, and evolving solid–fluid composition lie outside the one-dimensional representation. Repeat topography and deposit mapping, surveyed sections, additional arrival-time observations, and two-dimensional or multiphase calculations would provide stronger constraints on local flow intensity.

Meteorological, cryospheric, and seismic records distinguish immediate conditions from regional preconditioning but do not reveal the precise failure mechanism at the source. The absence of a nearby preceding earthquake and of an extreme short-duration meteorological anomaly weighs against those immediate triggers. Multi-decadal warming, rising glacier equilibrium-line altitude, reduced snowfall, and regional evidence of frozen-ground change are consistent with progressive weakening of an ice-bearing rock slope, while tectonic deformation and historical shaking provide a plausible fractured geological setting. None of these regional records measures fracture ice, rock temperature, water pressure, or displacement on the failed slope before 26 August 2026. Resolving the relative roles of

thermal change, freeze–thaw cycling, changing glacier support, and inherited damage would require source-scale observations and repeated deformation measurements.

7 Conclusion

The 26 August 2026 cascade began with an ice–rock slide on the north flank of the Langtang Lirung massif. The mapped source comprises approximately 12.6 Mm^3 of ice and 90.75 Mm^3 of rock and a combined volume of approximately 103 Mm^3. After striking a detached glacier tongue and its supraglacial lake, the slide developed into an ice–rock avalanche. Approximately 67 Mm^3 accumulated in Chhochen Khola, while about 36 Mm^3 entered the Lende Khola as a debris flow.

The three-stage cascade travelled approximately 22.32 km to Gyirong Port in about 8 min, at an overall mean velocity of approximately 46.5 m s^{-1}. Timing-derived mean velocities decreased from approximately 66.0 to 61.5 and 40.5 m s^{-1} across the ice–rock slide, ice–rock avalanche, and debris flow as mean gradients decreased from 58.8% to 17.9% and 7.8%. The Stage 1 Voellmy calculation yielded a local maximum above 100 m s^{-1}. At a bedrock gorge near the port, peak discharge was estimated at 2.46×10^5 m^3 s^{-1} with a cross-sectional mean velocity of 36.2 m s^{-1}.

One-dimensional Saint-Venant routing gives an S10 peak discharge of 2.408×10^5 m^3 s^{-1}, approximately 8.3% of the 2.914×10^6 m^3 s^{-1} confluence input. W50 increases from 10.96 to 111.47 s, documenting peak attenuation and hydrograph broadening downstream. The modeled S1–S10 peak-propagation velocity of 48.97 m s^{-1} exceeds the observed estimate of 40.5 m s^{-1} by about 17.3%. Source composition, glacier impact, intermediate storage, and channel routing jointly shaped the flow that reached Gyirong Port.

No intense rainfall, extreme short-term temperature anomaly, or nearby preceding M 4.5 or larger earthquake was identified as the immediate trigger. Multi-decadal warming, glacier retreat, freeze–thaw cycling, and geological preconditioning provide a plausible setting for progressive source weakening, although their individual contributions remain

unresolved. The event shows that source instability, slope breaks, intermediate deposition, water acquisition, and downstream flow transformation all matter to Himalayan cascade hazards. Stage-resolved analysis, rapid seismic detection, precomputed routing scenarios, and transboundary communication are therefore important for hazard assessment and warning in glacierized valleys.

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Author Contribution

LI Hao: Methodology, Field investigation, Formal analysis, Writing –original draft. LI Pu: Methodology, Formal analysis, Writing –original draft. HU Kaiheng: Project administration, Conceptualization, Methodology, Field investigation, Funding acquisition, Writing –original draft, Writing – review & editing, Supervision. LIU Junyu: Methodology, Formal analysis, Writing –original draft. LIU Shuang: Field investigation, Formal analysis, Writing – review & editing. KANG Shichang: Field investigation, Formal analysis, Writing – review & editing. SU Pengcheng: Field investigation, Writing – review & editing. YOU Yong: Formal analysis, Writing – review & editing.

Author Declaration

Availability of Data/Materials: The datasets generated during this study are available from the corresponding author upon reasonable request.

Conflicts of Interest: The 3rd author, HU Kaiheng, and the 6th author, KANG Shichang, are the editorial board member in Journal of Mountain Science. They were not involved in the journal's review of, or decisions related to, this manuscript. The authors declare no conflict of interest.

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